



Bluestock Mutual Fund Analytics Capstone Project

Internship Report

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Executive Summary

The Bluestock Mutual Fund Analytics Capstone project was undertaken to develop an end-to-end analytics solution for analysing the Indian mutual fund industry. The project integrated multiple datasets covering mutual fund schemes, NAV history, Assets Under Management (AUM), SIP inflows, investor transactions, benchmark indices, portfolio holdings, and fund performance metrics. A structured data pipeline was established using Python and SQLite to support data ingestion, cleaning, transformation, validation, and storage. This framework enabled the conversion of large volumes of raw financial data into reliable and analysis-ready datasets for further exploration and reporting.

Comprehensive exploratory and performance analytics were performed to study industry growth, investor behaviour, portfolio allocation patterns, category-wise fund flows, and scheme performance. Key risk-adjusted metrics including Sharpe Ratio, Sortino Ratio, Alpha, Beta, CAGR, Tracking Error, and Maximum Drawdown were calculated to evaluate fund performance under varying market conditions. The final output was delivered through interactive Tableau dashboards that provide insights into market trends, fund performance, investor participation, and portfolio characteristics, enabling stakeholders to make informed and data-driven investment decisions.

Table 1: Technology Stack Used in the Project

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Database	SQLite
Query Language	SQL
Exploratory Data Analysis	Jupyter Notebook
Performance Analytics	Python Statistical Libraries
Data Visualisation	Tableau
Version Control	Git
Documentation	LaTeX

Chapter 1

Data

The Bluestock Mutual Fund Analytics Capstone project utilised multiple datasets representing different dimensions of the Indian mutual fund ecosystem. These datasets collectively provide information related to mutual fund schemes, NAV movements, investor participation, portfolio allocation, benchmark performance and industry growth trends. The integration of these datasets enabled a comprehensive analytical framework for evaluating fund performance, investor behaviour and market dynamics.

The Fund Master dataset served as the primary reference table containing scheme-level attributes such as fund house, category, benchmark, expense ratio, risk classification and fund manager information. NAV History data provided daily valuation records which formed the basis for return calculations and trend analysis. Industry datasets including AUM, SIP inflows and folio counts were used to analyse mutual fund industry growth and investor participation trends.

Table 1.1: Summary of Major Datasets Used

Dataset	Records	Purpose
Fund Master	40	Scheme metadata and classification
NAV History	46,000	NAV trend and return analysis
AUM by Fund House	90	Industry growth analysis
Monthly SIP Inflows	48	SIP participation trends
Category Inflows	144	Category-wise investment flows
Industry Folio Count	21	Investor participation analysis
Scheme Performance	40	Risk-return evaluation
Investor Transactions	32,778	Investor behaviour analysis
Portfolio Holdings	322	Portfolio composition analysis
Benchmark Indices	8,050	Market benchmark comparison

Collectively, the project processed more than 87,000 records spanning fund-level, investor-level, portfolio-level and market-level information. These datasets formed the foundation for ETL operations, exploratory data analysis, performance analytics and dashboard development. The availability of both transactional and market datasets enabled the project to perform comprehensive analyses ranging from investor behaviour assessment to risk-adjusted fund performance evaluation.

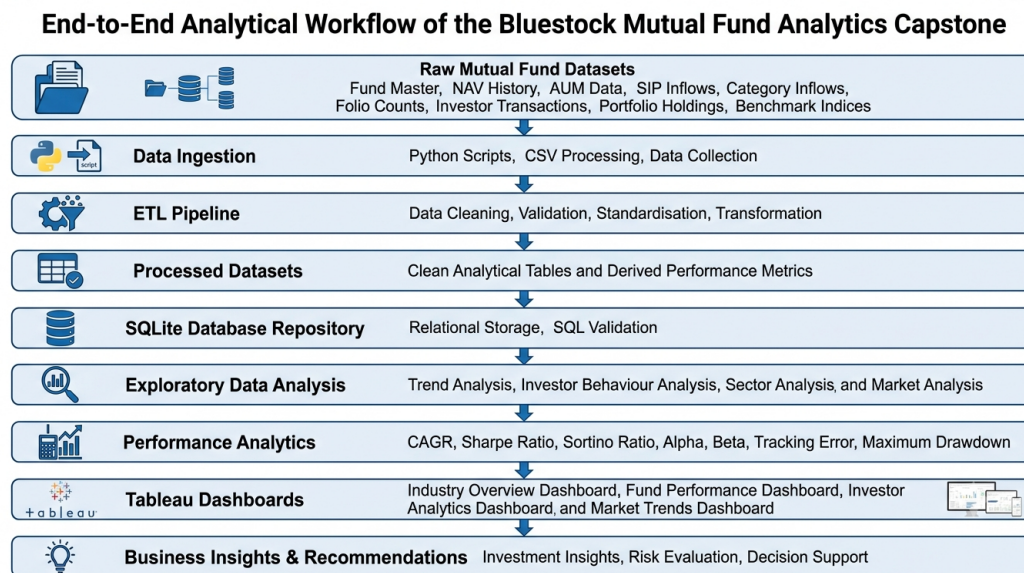


Figure 1.1: End-to-End Analytical Workflow of the Bluestock Mutual Fund Analytics Capstone

Figure 1.1 illustrates the overall analytical workflow adopted in the project. Raw mutual fund datasets were extracted from multiple sources and processed through a Python-based ETL pipeline involving data ingestion, cleaning, validation and transformation. These steps ensured data consistency, reliability and readiness for downstream analytical tasks.

The processed datasets were subsequently stored in a centralised SQLite repository that served as the primary analytical data source for the project. The repository supported exploratory data analysis, advanced performance calculations and dashboard development. Finally, the transformed information was integrated into Tableau dashboards to provide stakeholders with interactive insights related to fund performance, industry growth trends, investor participation and portfolio composition.

Chapter 2

ETL Process

The ETL (Extract, Transform, Load) process was used to prepare data for analysis in the Bluestock Mutual Fund Analytics Capstone project. Data from multiple sources was processed through a structured pipeline to ensure consistency, quality, and reliability. Python-based scripts and notebooks were used to automate data preparation and transformation.

The extraction phase involved loading raw CSV datasets containing mutual fund schemes, NAV history, AUM growth, SIP inflows, investor transactions, portfolio holdings, benchmark indices, and industry statistics. Basic data profiling was performed to verify dataset structure, datatypes, missing values, and schema consistency before transformation.

```
data_quality_report.txt data_ingestion_output.txt x schema.sql load_to_sqlite.py queries.sql dat
reports > data_ingestion_output.txt
55 =====
56 FILE: 02_nav_history.csv
57
58 SHAPE:
59 (46000, 3)
60
61 DATA TYPES:
62 amfi_code    int64
63 date         object
64 nav          float64
65 dtype: object
66
67 FIRST 5 ROWS:
68 | amfi_code    date         nav
69 | 119551      2022-01-03    54.3856
70 | 119551      2022-01-04    54.3474
71 | 119551      2022-01-05    54.6809
72 | 119551      2022-01-06    55.4558
73 | 119551      2022-01-07    55.3692
74 |
75 =====
76 FILE: 04_monthly_sip_inflows.csv
77
78 SHAPE:
79 (46, 6)
80
81 DATA TYPES:
82 month        object
83 sip_inflow_crore    int64
84 active_sip_accounts_crore    float64
85 new_sip_accounts_lakh    float64
86 sip_aum_lakh_crore    float64
87 yoy_growth_pct    float64
88 dtype: object
89
90 FIRST 5 ROWS:
91 | month sip_inflow_crore active_sip_accounts_crore new_sip_accounts_lakh sip_aum_lakh_crore yoy_growth_pct
92 | 2022-01 11517 4.91 9.10 4.88 NaN
93 | 2022-02 11438 4.93 8.28 4.85 NaN
94 | 2022-03 12328 5.49 18.58 5.01 NaN
95 | 2022-04 11863 5.48 9.52 5.12 NaN
96 | 2022-05 12286 5.55 8.11 5.15 NaN
```

Figure 2.1: Data Ingestion and Initial Profiling Output

The data ingestion process successfully loaded all source datasets into the analytical environment. Dataset dimensions, column structures, and sample records were validated to ensure completeness and consistency. Early profiling helped identify datatype

mismatches and formatting issues that were addressed during the transformation stage, improving overall data quality.

During transformation, data cleaning and standardisation procedures were performed using the Pandas library. Date fields were converted into consistent formats, categorical attributes were normalised, and numerical values were validated. Missing values, duplicate records, and data quality issues were handled to ensure that all datasets followed a uniform structure suitable for analysis.

Several derived datasets were generated to support advanced analytics, including return calculations, CAGR reports, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, Maximum Drawdown, and Fund Scorecards. These datasets enabled deeper performance and risk analysis across mutual fund schemes.

The final loading phase stored the cleaned and transformed datasets in a SQLite database. Tables were created using predefined schemas and validated through SQL queries. The database served as a central repository for data exploration, dashboard integration, and analytical reporting, ensuring consistency across all outputs.

date	fund_name	num_sahs_store	num_store	num_schemes
2022-03-31	SBF Mutual Fund	6.05	402000	188
2022-03-31	KCI Mutual Fund	4.60	402000	270
2022-03-31	HDFC Mutual Fund	4.35	435000	195
2022-03-31	Nippon India MF	2.7	270000	177
2022-03-31	Kotak Mahindra MF	2.7	270000	188
2022-03-31	Aditya Birla Sun Life MF	2.78	278000	189
2022-03-31	Axis Mutual Fund	2.5	250000	95
2022-03-31	UTI Mutual Fund	2.3	230000	142
2022-03-31	Mitras Asset MF	1.95	195000	56
2022-03-31	CSP Mutual Fund	1.1	110000	88
2022-09-30	SBF Mutual Fund	6.3	430000	188
2022-09-30	KCI Mutual Fund	4.84	484000	270
2022-09-30	HDFC Mutual Fund	4.40	440000	195
2022-09-30	Nippon India MF	2.78	278000	177
2022-09-30	Kotak Mahindra MF	2.72	272000	188
2022-09-30	Aditya Birla Sun Life MF	2.85	285000	189
2022-09-30	Axis Mutual Fund	2.4	240000	95
2022-09-30	UTI Mutual Fund	2.32	232000	142
2022-09-30	Mitras Asset MF	1.98	198000	56
2022-09-30	CSP Mutual Fund	1.12	112000	88
2023-03-31	SBF Mutual Fund	7.17	717000	188
2023-03-31	KCI Mutual Fund	5	500000	270
2023-03-31	HDFC Mutual Fund	4.5	450000	195
2023-03-31	Nippon India MF	2.83	283000	177
2023-03-31	Kotak Mahindra MF	2.85	285000	188
2023-03-31	Aditya Birla Sun Life MF	2.75	275000	189
2023-03-31	Axis Mutual Fund	2.41	241000	95
2023-03-31	UTI Mutual Fund	2.38	238000	142
2023-03-31	Mitras Asset MF	1.16	116000	56
2023-03-31	CSP Mutual Fund	1.15	115000	88

Figure 2.2: SQLite Database Structure and Analytical Repository

The SQLite repository consolidated all processed datasets into a structured environment suitable for querying and reporting. The database design simplified data validation and ensured efficient retrieval of information required for dashboard visualisations and performance analytics. The use of SQL-based validation further strengthened confidence in the quality and consistency of the transformed data.

SQL validation queries were executed to verify record counts, primary key uniqueness, data consistency and aggregation results. Validation checks ensured that transformed datasets accurately reflected the source information before analytical processing.

Overall, the ETL pipeline successfully converted raw financial and investor datasets into a reliable analytical repository.

Chapter 3

Exploratory Data Analysis

EDA was conducted to understand industry growth patterns, fund performance behaviour and investor participation trends. Multiple visualisations were generated to identify key patterns across the datasets.

3.1 NAV Trend Analysis

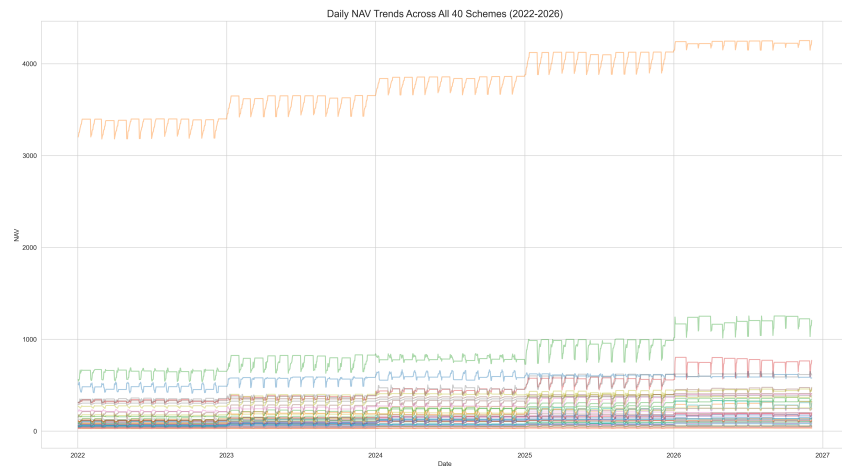


Figure 3.1: NAV Trend Analysis

The NAV trend chart illustrates the movement of scheme values over time. The analysis highlights the long-term growth trajectory observed across major equity-oriented mutual funds. Consistent upward movement indicates wealth creation potential while temporary fluctuations reflect market corrections and volatility periods. Understanding NAV behaviour is important for assessing historical performance and investment stability.

3.2 AUM Growth by AMC

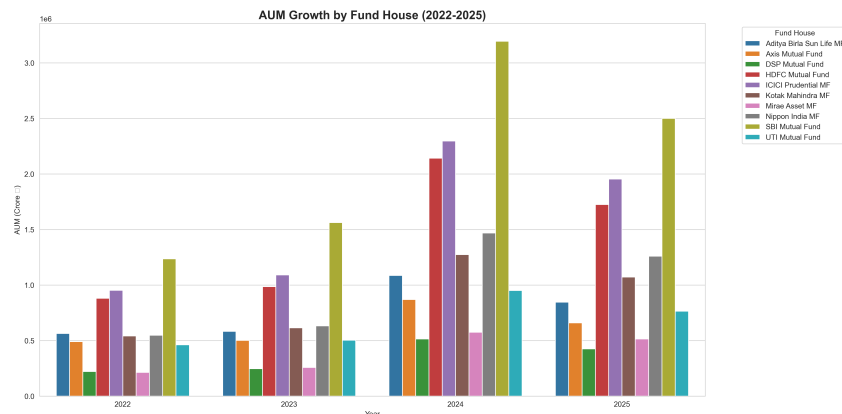


Figure 3.2: AUM Growth by Fund House

Assets Under Management provide an indication of investor confidence and market presence. The visualisation compares major fund houses and reveals the relative scale at which different AMCs operate. Larger AUM values generally indicate stronger distribution networks, broader investor participation and higher institutional trust.

3.3 SIP Inflow Trends

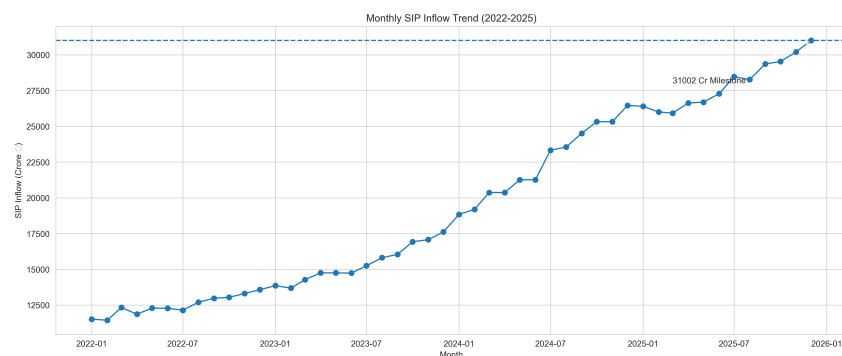


Figure 3.3: SIP Inflow Trend

Systematic Investment Plans continue to be a major driver of mutual fund growth in India. The chart demonstrates increasing SIP inflows over time and reflects growing retail participation. Rising SIP adoption indicates investor preference for disciplined investing and long-term wealth creation strategies.

3.4 Category Heatmap



Figure 3.4: Category Inflow Heatmap

The heatmap identifies category-wise inflow and outflow patterns. Certain categories consistently attract capital while others experience cyclical movements. This analysis assists fund managers in understanding investor preferences and evolving market sentiment.

3.5 Sector Allocation

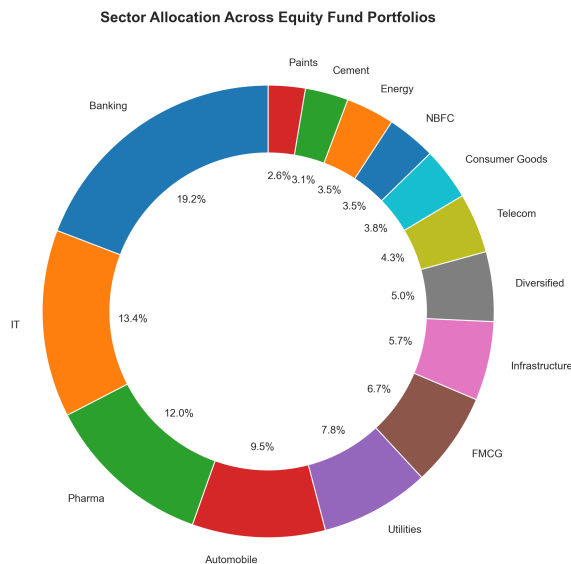


Figure 3.5: Sector Allocation

Sector allocation analysis reveals portfolio diversification strategies followed by mutual funds. The chart identifies sectors receiving significant exposure and highlights investment concentration trends. Diversified allocation patterns generally contribute to better risk management.

Chapter 4

Findings

The analysis generated several important insights regarding investor behaviour, industry growth, and mutual fund performance. Results obtained from NAV history, SIP inflows, AUM data, folio statistics, portfolio holdings, and investor transactions indicate continued expansion of the Indian mutual fund industry.

The findings highlight increasing retail participation, strong growth in systematic investments, and the dominance of established asset management companies. At the same time, expanding investor penetration and diversified portfolio allocations suggest a maturing investment ecosystem supported by growing financial awareness.

4.1 Growth in SIP Participation

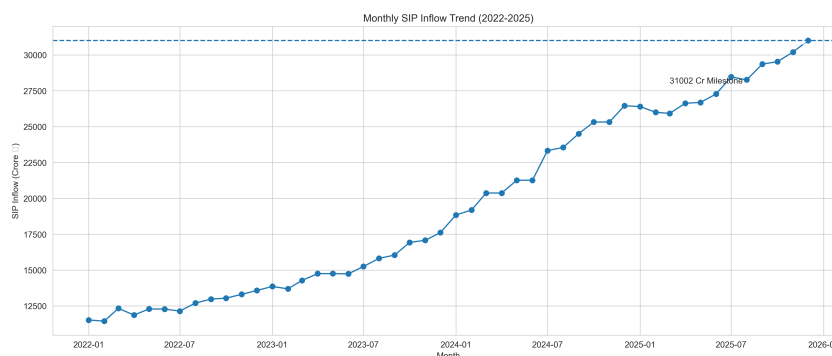


Figure 4.1: SIP Inflow Trend

SIP inflows have shown a consistent upward trend, reflecting growing investor confidence in systematic investing. The results indicate increasing adoption of disciplined investment strategies and greater awareness of long-term wealth creation opportunities.

Retail investors continue to view SIPs as an accessible and effective investment vehicle, contributing significantly to overall industry growth.

4.2 Increase in Investor Participation

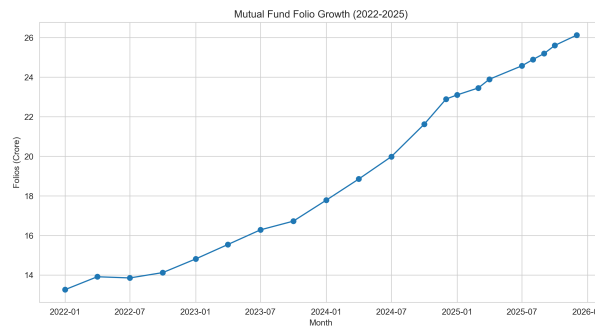


Figure 4.2: Industry Folio Growth

The steady rise in folio counts demonstrates expanding participation across the mutual fund industry. Increasing numbers of investor accounts indicate broader market penetration and improved accessibility through digital investment platforms.

The trend also reflects growing financial literacy and rising acceptance of market-linked investment products among retail investors.

4.3 Dominance of Major Fund Houses

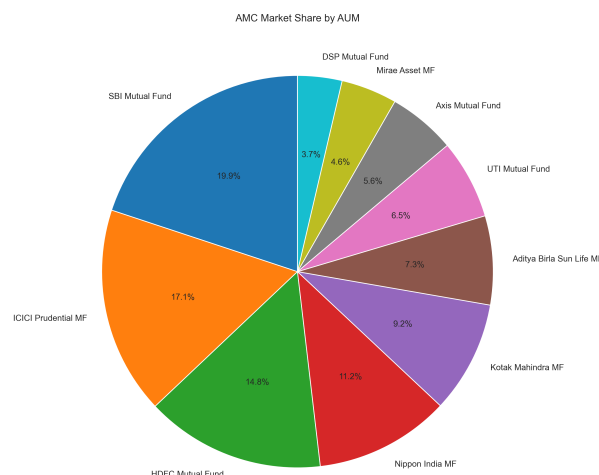


Figure 4.3: AMC Market Share

Market share analysis reveals that a small number of large asset management companies continue to manage a significant portion of industry assets. Their strong distribution networks, brand recognition, and extensive product offerings contribute to their leadership positions. At the same time, the presence of smaller players indicates healthy competition and continued opportunities for growth within the industry.

Chapter 5

Performance Analysis

Risk-adjusted performance metrics were computed to evaluate mutual fund efficiency. The analysis incorporated CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error and Maximum Drawdown calculations. These measures provide a balanced assessment of both return generation and portfolio risk, enabling a more comprehensive evaluation of fund performance.

5.1 Rolling Sharpe Analysis

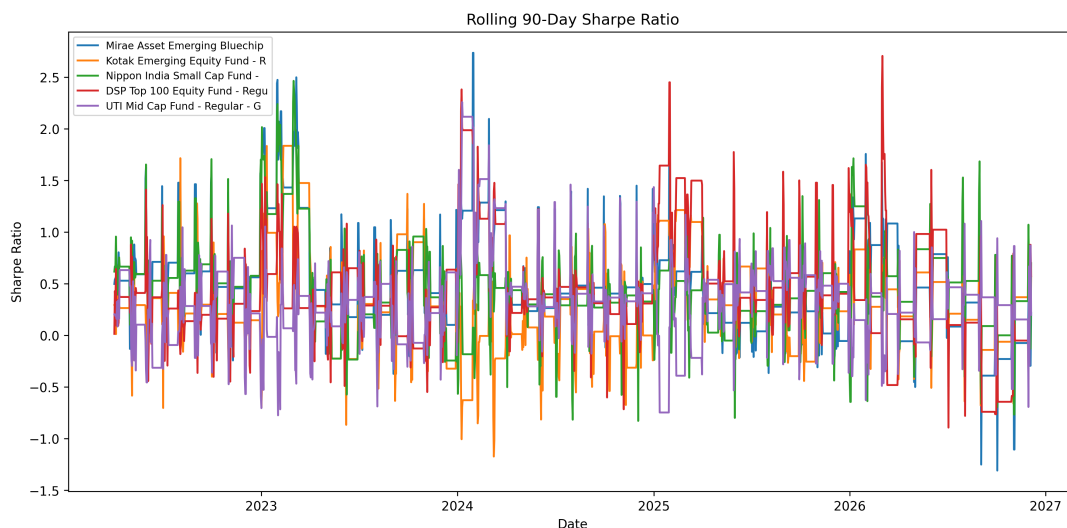


Figure 5.1: Rolling Sharpe Ratio

The rolling Sharpe Ratio measures excess return generated per unit of risk. Higher values indicate superior risk-adjusted performance. The chart highlights periods of strong performance and demonstrates how market conditions influence portfolio efficiency over time.

Variations in the ratio reflect changes in return consistency and market volatility across different periods. Monitoring rolling Sharpe values helps identify phases where fund performance improved or deteriorated relative to the level of risk undertaken.

5.2 Sector Concentration Analysis

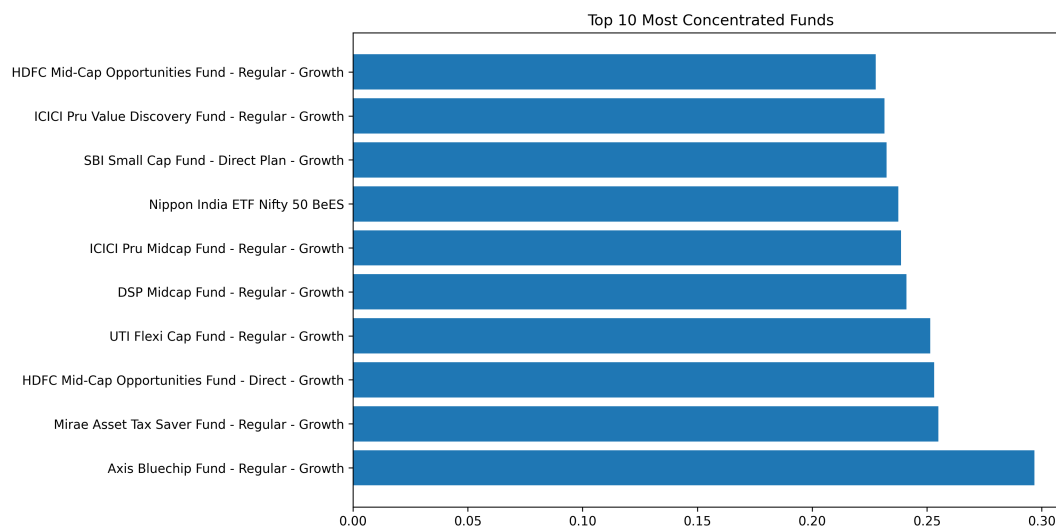


Figure 5.2: Sector Concentration Analysis

The Herfindahl-Hirschman Index was used to evaluate portfolio concentration. Lower concentration values indicate better diversification while higher values suggest increased dependence on specific sectors. This analysis supports risk monitoring and portfolio construction decisions.

The results provide insight into how effectively investments are distributed across sectors within the portfolio. Maintaining appropriate diversification can reduce exposure to sector-specific risks and improve long-term portfolio stability.

The performance evaluation framework provides a comprehensive view of both return generation and risk exposure. Combining multiple metrics enables more reliable assessment than relying solely on absolute returns. The findings support informed investment decision-making by highlighting the relationship between portfolio returns, volatility, and diversification characteristics.

Chapter 6

Interactive Dashboards

Interactive dashboards were developed using Tableau to provide stakeholders with an intuitive and data-driven view of mutual fund industry trends, fund performance, investor behaviour, and market dynamics. The dashboards support exploratory analysis through visualisations, filters, and performance indicators, enabling users to derive insights efficiently.

6.1 Industry Overview Dashboard

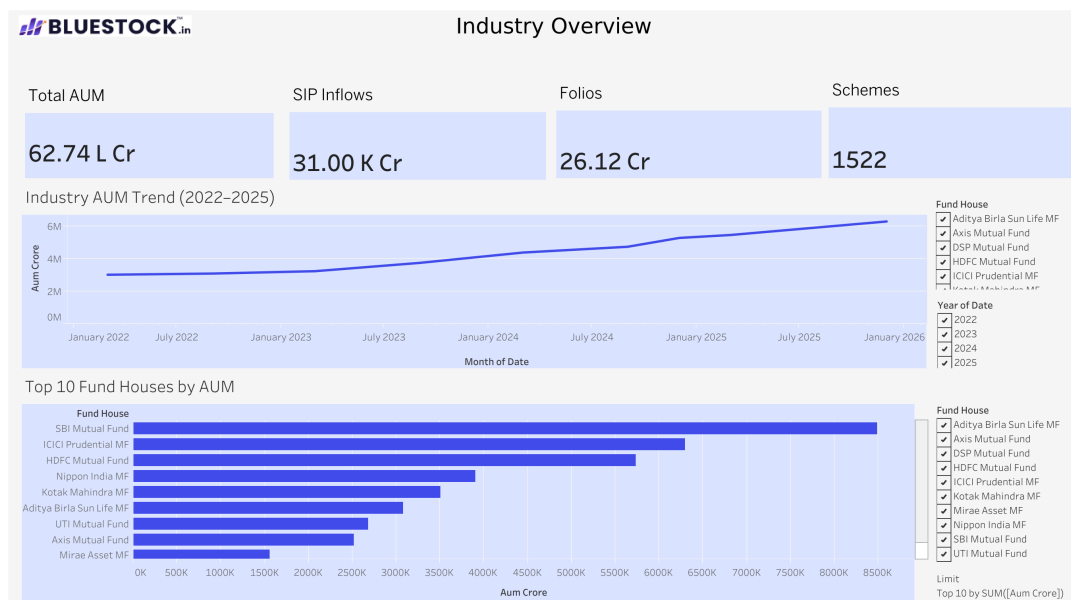


Figure 6.1: Industry Overview Dashboard

This dashboard presents a consolidated view of the mutual fund industry including AUM, SIP inflows and investor participation metrics. It serves as the primary entry point for stakeholders seeking a high-level overview of industry performance.

6.2 Fund Performance Dashboard

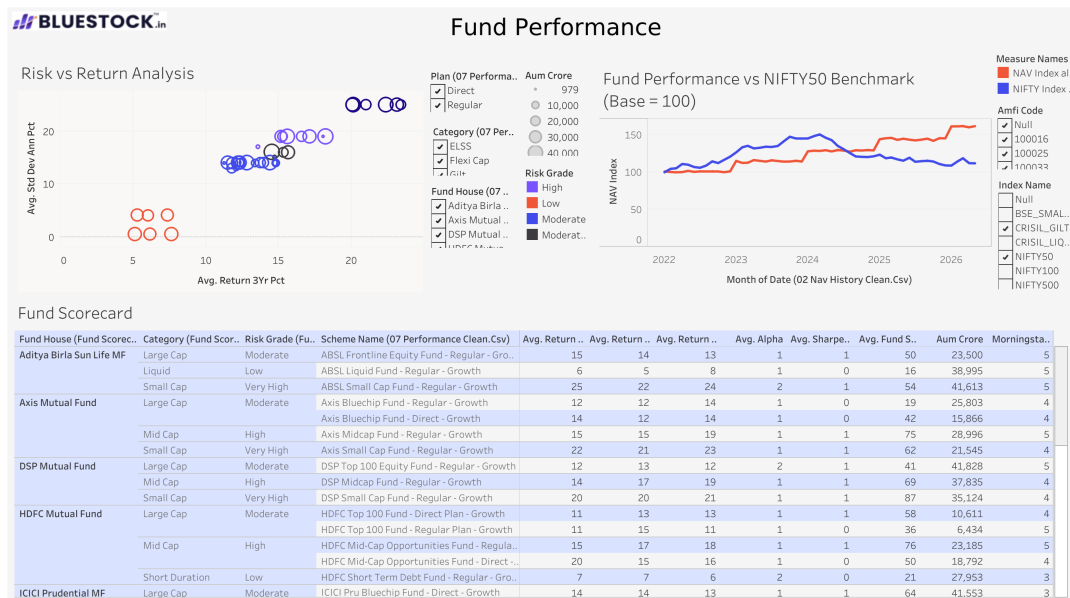


Figure 6.2: Fund Performance Dashboard

The performance dashboard enables comparison of schemes using return and risk measures. Interactive filtering allows users to identify top-performing funds and evaluate their consistency over time.

6.3 Investor Analytics Dashboard

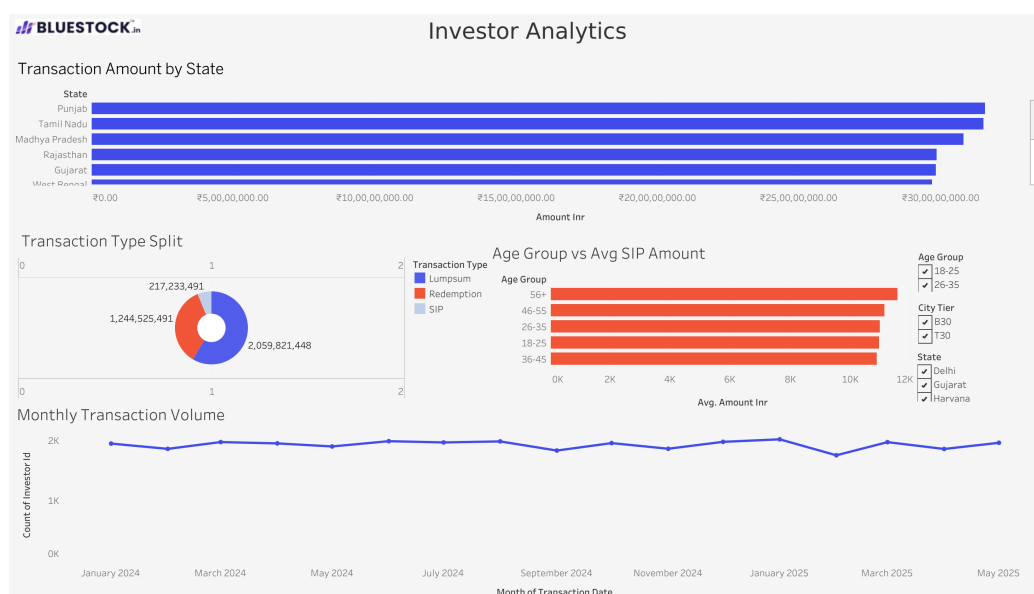


Figure 6.3: Investor Analytics Dashboard

This dashboard focuses on demographic trends and transaction behaviour. Users can analyse investment activity across age groups, geographic regions and transaction categories.

6.4 Market Trends Dashboard

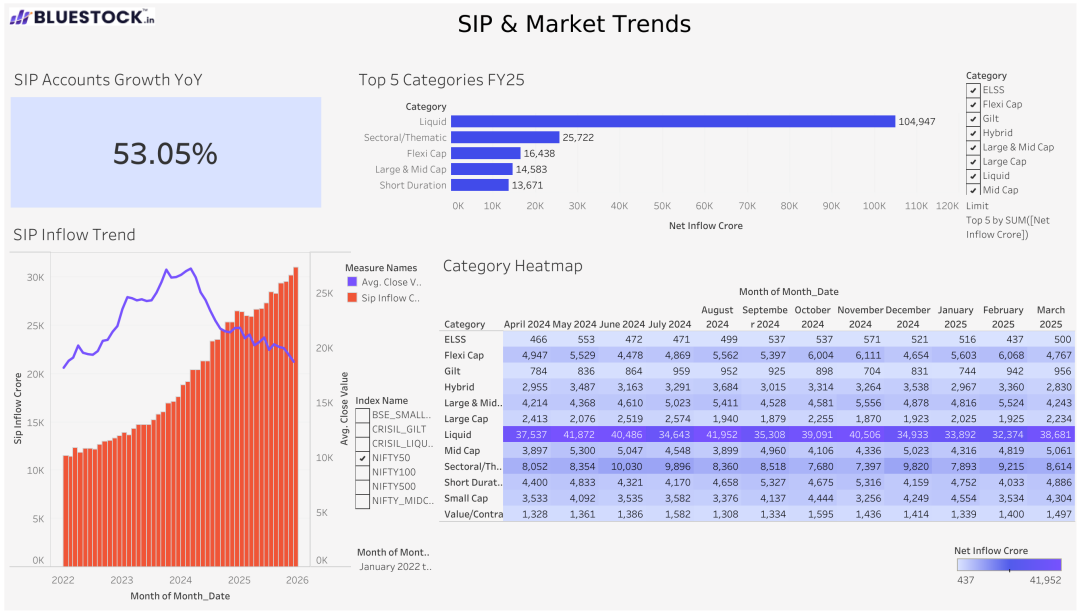


Figure 6.4: Market Trends Dashboard

The market trends dashboard highlights SIP growth, category flows and industry momentum. It supports strategic planning and provides insights into changing investor preferences.

Trend-based visualisations enable users to monitor market developments and evaluate the evolving investment landscape.

Chapter 7

Recommendations

The project findings suggest several opportunities for improvement. Asset management companies should continue promoting SIP-based investment strategies due to their strong contribution to industry growth. Investor education initiatives can further increase participation in emerging regions and underrepresented demographic groups.

Risk-adjusted metrics should be integrated into routine performance monitoring frameworks to support informed decision making. Portfolio diversification should remain a priority to reduce sector-specific concentration risks. Additionally, advanced analytics and predictive modelling can be incorporated into future dashboard versions to improve forecasting capabilities.

Asset management companies may also leverage investor demographic and transaction data to develop more targeted products and personalised engagement strategies. Enhanced digital platforms, combined with data-driven insights, can improve customer experience and strengthen long-term investor retention. Continued investment in analytics capabilities will help organisations respond more effectively to evolving market conditions and investor preferences.

Future enhancements to this project may include the integration of machine learning models for return forecasting, investor segmentation, and fund recommendation systems. Real-time data integration, automated alert mechanisms, and predictive analytics could further enhance the decision-making capabilities of the platform. These improvements would provide deeper insights to both investors and fund managers while increasing the overall value of the analytical framework.

Chapter 8

Limitations

The project relied on available historical datasets and simulated analytical environments. Certain market events, macroeconomic indicators and behavioural factors were not explicitly incorporated into the analysis. Results may vary as new market data becomes available.

The analysis primarily focused on selected schemes and fund houses included within the project scope. Real-world deployment would benefit from broader coverage, live data integration and additional validation using production-grade datasets.

Data quality and completeness were dependent on the accuracy of the source datasets. Although extensive cleaning and validation procedures were performed, inconsistencies or reporting differences across data sources may influence certain analytical outcomes. Such limitations are common when integrating data from multiple providers.

In addition, the dashboards and analytical models were designed primarily for educational and research purposes. While the generated insights are valuable for understanding industry trends and fund performance, they should not be considered as direct investment advice. Future enhancements involving real-time market data, larger datasets and advanced predictive models could further improve the accuracy and practical applicability of the platform.

Although the current framework successfully supports data-driven analysis, incorporating advanced capabilities in future iterations could enhance analytical depth, predictive accuracy, and overall decision-making effectiveness.

Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully demonstrated the complete lifecycle of a modern data analytics project, beginning with data acquisition and ending with business intelligence delivery through interactive dashboards. The project integrated multiple mutual fund datasets and established a structured analytical framework involving data ingestion, cleaning, transformation, validation, exploratory analysis, performance evaluation, and dashboard development.

The analysis revealed several important trends within the Indian mutual fund industry, including the steady growth of SIP investments, increasing investor participation, expansion across geographic regions, and the continued market leadership of major asset management companies. Investor demographic analysis provided insights into participation patterns across age groups, income segments, and city tiers, while portfolio and sector allocation studies highlighted diversified characteristics across mutual fund schemes.

A comprehensive performance evaluation framework was developed using key risk-adjusted metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, CAGR, and Maximum Drawdown. These measures enabled a balanced assessment of both returns and associated risks, allowing meaningful comparisons across schemes and benchmark indices.

From a business perspective, the solution delivers a centralised platform for monitoring mutual fund performance, analysing industry growth trends, evaluating portfolio characteristics, and understanding investor behaviour. The interactive Tableau dashboards provide an intuitive interface for exploring complex financial data and generating actionable insights, thereby supporting data-driven investment analysis and strategic decision-making.

The final solution establishes a scalable foundation for future enhancements such as predictive analytics, recommendation systems, real-time NAV integration, portfolio optimisation models, and advanced risk forecasting. Overall, the project demonstrates how data engineering, financial analytics, and business intelligence techniques can be integrated to create a comprehensive mutual fund analytics ecosystem capable of delivering practical value to investors, analysts, and financial institutions.